



K L Deemed to be University

Department of computer science and Engineering -1 -- KLVZA
Course Handout
2025-2026, Term3

Course Title	: DATA STRUCTURES AND ALGORITHMS - 2
Course Code	: 25SC1305E
L-T-P-S Structure	: 2-0-2-4
Pre-requisite	:
Credits	: 4
Course Coordinator	: Zeelan Basha CMAK
Team of Instructors	:
Teaching Associates	:
Syllabus :	BST concepts and operations, tree traversals, complexity analysis, AVL trees and rotations, AVL insertion and deletion, BST vs AVL comparison, B-Tree and B+ Tree concepts and operations, database indexing, segment trees and range queries, lazy propagation, Fenwick trees (BIT), tree comparison techniques, graph fundamentals and representations, BFS and DFS traversals, connected components and union-find, minimum spanning trees (Kruskal's and Prim's), shortest path algorithms (Dijkstra, Bellman-Ford, Floyd-Warshall), topological sorting and strongly connected components, divide and conquer paradigm, merge sort and quick sort, heap sort, counting and radix sort, sorting performance and benchmarking, greedy algorithms and design strategies, activity selection and fractional knapsack, Huffman coding and coin change, dynamic programming concepts, memoization and tabulation, 0/1 knapsack, longest common subsequence, longest increasing subsequence, matrix chain multiplication, edit distance, greedy vs dynamic programming decision making
Text Books :	Introduction to Algorithms (CLRS), 4th Edition Algorithms, 4th Edition Data Structures and Algorithm Analysis in Java, 3rd Edition Algorithm Design
Reference Books :	Algorithms Unlocked Grokking Algorithms, 2nd Edition Algorithms Unlocked
MOOCS :	Algorithms, Part I & Part II Coursera — Princeton University (Sedgewick & Wayne) Data Structures and Algorithms Specialization Coursera — UC San Diego (Pevzner, Levin) Data Structures & Algorithms Nanodegree Udacity — Udacity Instructors Algorithms and Data Structures edX — MIT / Microsoft (various) Data Structures and Algorithms using Java NPTEL — IIT Madras
Course Rationale :	Building on the foundational data structures in DSA-I, this course advances students into tree structures, graph algorithms, advanced sorting paradigms, and dynamic programming. Students encounter real-world problem scenarios in each module, learning to select the correct data structure and algorithmic strategy based on problem constraints, time complexity requirements, and industry usage — preparing them for competitive programming, technical interviews, and software engineering roles.
Course Objectives :	Building on the foundational data structures in DSA-I, this course advances students into tree structures, graph algorithms, advanced sorting paradigms, and dynamic programming. Students encounter real-world problem scenarios in each module, learning to select the correct data structure and algorithmic strategy based on problem constraints, time complexity requirements, and industry usage — preparing them for competitive programming, technical interviews, and software engineering roles. 1. Design and implement tree-based data structures including BST, AVL, B-Trees, and Segment Trees for hierarchical data problems. 2. Apply graph traversal and shortest-path algorithms to solve network routing, social network analysis, and infrastructure planning problems. 3. Master advanced divide-and-conquer sorting and selection algorithms and analyze their complexity with mathematical rigor. 4. Formulate greedy and dynamic programming solutions for optimization problems arising in scheduling, resource allocation, and sequence analysis.

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Apply Binary Search Trees (BST) and self-balancing AVL Trees to construct, insert, delete, and search hierarchical datasets, demonstrating correct rotation operations and $O(\log n)$ performance guarantees. Apply multi-way search trees (B-Trees, B+ Trees) .	PO2,PO3,PO4,PO5,PO7,PO1	4
CO2	Apply Range Query structures (Segment Trees, Fenwick Trees) to select appropriate structures for database indexing, storage systems, and interval query problems.BFS and DFS graph traversal algorithms and construct Minimum Spanning Trees using Kruskal's and Prim's algorithms to solve infrastructure design, network connectivity, and clustering problems in weighted undirected graphs.	PO3,PO9,PO7,PO5,PO4,PO2,PO1	4
CO3	Analyze and implement Dijkstra's, Bellman-Ford, and Floyd-Warshall algorithms to compare their applicability and correctness across different graph types (weighted/unweighted, directed/undirected, negative weights) in real-world routing and logistics systems.Evaluate and implement sorting algorithms (Merge Sort, Quick Sort)	PO11,PO10,PO9,PO7,PO4,PO3,PO2,PO1,PO5	4
CO4	Evaluate and implement advanced sorting algorithms (Heap Sort, Counting Sort, Radix Sort) and design optimal solutions using Greedy and Dynamic Programming techniques (Knapsack, LCS, Matrix Chain Multiplication), justifying choices based on time complexity, space usage, stability, and problem characteristics.	PO4,PO10,PO9,PO7,PO5,PO3,PO2,PO1	4
CO5	Design, Develop and evaluate common practical applications for Non linear Data Structures,Dynamic programming,and Greedy Algorithms	PO1,PO11,PO9,PO5,PO4,PO3,PO2	5
CO6	Skill the students in such a way that students will be able to develop and create programs as well as applications in non linear DS	PO1,PO2,PO3,PO5,PO9,PO11	6

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4	COI-5	COI-6
CO1	4	Btl-1 Remember BST concepts, properties, tree traversals	Btl-2 Understand Complexity analysis of tree operations	Btl-3 Apply AVL trees, rotations, insertion and deletion	Btl-4 Compare and analyze the concepts of BST vs AVL trees,Remember the concepts of B-Tree and B+ Tree concepts		
CO2	4	Btl-1 Remember the concepts of	Btl-2 Understand the priority queues concept and	Btl-3 Apply the concepts of Segment trees, range queries, lazy propagation,concepts	Btl-4 Analyze the concepts of Fenwick trees (BIT) and compare tree		

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4	COI-5	COI-6
		Segment Trees, Fenwick Trees Tree concepts	Database indexing using trees, Graph fundamentals and representations	of BFS and DFS traversals	structures, Connected components, Union-Find, MST (Kruskal's & Prim's)		
CO3	4	Btl-1 Remember the concept of Shortest path algorithms,	Btl-2 Understand the concepts of Bellman-ford, floyd-warshall algorithms	Btl-3 Apply the concepts of different shortest routing algorithms by comparing different graph types.	Btl-4 Analyze the concepts of Shortest path algorithms, Topological sorting, SCC, Merge Sort, Quick Sort		
CO4	4	Btl-1 Remember the concepts of Heap, Counting sorts	Btl-2 Understand the concepts of advance sorting techniques - Radix Sort	Btl-3 Apply Sorting techniques implementation & benchmarking	Btl-4 Analyze Greedy algorithms (Activity selection, Knapsack, Huffman, Coin change)		
CO5	5	Btl-1 Remember Non-linear data structures concepts	Btl-2 Understand Applications of non-linear data structures	Btl-3 Apply the non-linear data structures in real-time scenarios	Btl-4 Analyze the Differentiate structures based on properties and performance	Btl-5 Evaluate suitability of non-linear data structures for real-world problems	
CO6	6	Btl-1 Recall the non linear data structures	Btl-2 Understand non linear data structures with real time applications	Btl-3 Use the non linear data structures with real time applications	Btl-4 Analyze Edit distance and optimization problem		Btl-5 Evaluate Compare Greedy vs Dynamic Programming and choose optimal approach

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

Po No.	Program Outcome
PO5	Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO9	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-1	Binary Search Trees (BST)	Ref Book [5],CH 2Page no 12-22	Chalk,Talk	End Semester Exam,Global Challenges,SEM-EXAM1
2	CO1	COI-2	Binary Search Tree (BST) Operations and analyzing the time complexity	Ref Book [5],CH 2Page no 15-22	Chalk,Talk	End Semester Exam,Hackathon-Final Review,SEM-EXAM1
3	CO1	COI-2	Binary Search Tree (BST) Operations and analyzing the time complexity.	Ref Book [5],CH 3Page no 28 -31	Chalk,Talk	End Semester Exam,SEM-EXAM1
4	CO1	COI-3	Self-balancing AVL Trees	Ref Book [3],CH 3Page no 93 -95	Chalk,Talk	End Semester Exam,SEM-EXAM1
5	CO1	COI-4	Self-balancing AVL Trees, comparing performance with Binary Search Tree.	Ref Book [2],CH 2Page no 260-269	Chalk,Talk	End Semester Exam,SEM-EXAM1
6	CO1	COI-4	Multi-way search trees (B-Trees)	Ref Book [2],CH 7.7Page no 270-287	Chalk,Talk	End Semester Exam,SEM-EXAM1
7	CO1	COI-4	Multi-way search trees (B+-Trees)	Ref Book [2],CH2Page no 288-307	Chalk,Talk	ALM,End Semester Exam,SEM-EXAM1
8	CO2	COI-1	Range Query structures (Segment Trees)	Ref Book [3],CH 7.6Page no 246 -250	Chalk,Talk	End Semester Exam,Global Challenges,SEM-EXAM1
9	CO2	COI-1	Range Query structures (Fenwick Trees)	Ref Book [3],CH 7.6Page no 246 -250	Chalk,Talk	End Semester Exam,Hackathon-Final Review,SEM-EXAM1
10	CO2	COI-	Graphs and terminology.	Ref Book [3],CH	Chalk,Talk	End Semester Exam,SEM-

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
		2	Implementing graph representations using adjacency matrix and adjacency list and analyzing space and time complexity	3Page no 188-192		EXAM1
11	CO2	COI-3	BFS graph traversal algorithm	Ref Book [3],CH 3Page no 188-192	Chalk,Talk	End Semester Exam,SEM-EXAM1
12	CO2	COI-3	DFS graph traversal algorithm	Ref Book [3],CH 3Page no 193-196	Chalk,Talk	End Semester Exam,SEM-EXAM1
13	CO2	COI-4	Construction of Minimum Spanning Trees using Kruskal's and Prim's algorithms	Ref Book [3],CH3Page no 193-196	Chalk,Talk	ALM,End Semester Exam,SEM-EXAM1
14	CO3	COI-1	Dijkstra's algorithm	Ref Book [3],CH 3Page no 203-205	Chalk,Talk	End Semester Exam,Global Challenges,SEM-EXAM2
15	CO3	COI-2	Priority Queues	Ref Book [3],CH 3Page no 206-209	Chalk,Talk	End Semester Exam,Hackathon-Final Review,SEM-EXAM2
16	CO3	COI-2	Bellman-Ford algorithm	Ref Book [3],CH 4Page no 151-166	Chalk,Talk	End Semester Exam,SEM-EXAM2
17	CO3	COI-2	Floyd-Warshall algorithm	Ref Book [3],CH 3Page no 136-142	Chalk,Talk	End Semester Exam,SEM-EXAM2
18	CO3	COI-4	Divide and Conquer strategy. Implementing Merge Sort and Quick Sort including partitioning and merging process	Ref Book [3],CH4Page no 145-149	Chalk,Talk	ALM,End Semester Exam,SEM-EXAM2
19	CO4	COI-1	Implementing Heap Sort and Quick Sort and analyzing performance for different input cases	Ref Book [3],CH 4Page no 143-144	Chalk,Talk	End Semester Exam,Global Challenges,SEM-EXAM2
20	CO4	COI-2	Implementing Counting Sort and Radix Sort and analyzing time complexity and limitations	Ref Book [3],CH 12Page no 579-587	Chalk,Talk	End Semester Exam,Hackathon-Final Review,SEM-EXAM2
21	CO4	COI-3	Empirical benchmarking of algorithms. Implementing all six sorting algorithms and running them on different datasets	Ref Book [1],CH 3.4Page no 95-101	Chalk,Talk	End Semester Exam,SEM-EXAM2
22	CO4	COI-4	Greedy approach and its properties. Implementing Activity Selection and Fractional Knapsack algorithms and analyzing correctness and efficiency of greedy strategy	Ref Book [2],CH 3Page no 458-485	Chalk,Talk	End Semester Exam,SEM-EXAM2

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
23	CO4	COI-4	Huffman Coding and greedy strategy for compression.	Ref Book [2],CH 3Page no 458 -485	Chalk,Talk	End Semester Exam,SEM-EXAM2
24	CO4	COI-4	Dynamic Programming, Implementing 0/1 Knapsack, Longest Common Subsequence, Longest Increasing Subsequence and Matrix Chain Multiplication	Ref Book [2],CH3Page no 458 -485	Chalk,Talk	ALM,End Semester Exam,SEM-EXAM2

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Students come to know about Binary Search Tree and its operations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Introducing Data Structures and Algorithms - II Syllabus	3	Chalk	--- NOT APPLICABLE ---
40	Introduce MediFlow – Smart Healthcare Management System to students as a real-time application of Data Structures. Explain that how the system manages patient records across multiple hospital branches, enabling doctors to efficiently access and update data from any location. Highlight how the project is suitable for different data structures. Introduce the list of nonlinear structures and brief how they are suitable for the project.	3	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	3	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 Students come to know about Binary Search Tree and its operations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	1	Chalk	--- NOT APPLICABLE ---
10	Discuss MediFlow, where fast retrieval and deletion of patient records is required, by explaining efficient operations on stored data	3	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of Binary Search Tree. Implementing insertion and tree traversals (inorder, preorder, postorder) and analyzing the time complexity.	3	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	3	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 1 Students come to know about Binary Search Tree and its operations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	3	Chalk	--- NOT APPLICABLE ---
10	Discuss MediFlow: How to search, delete, and display patient records, by explaining the need for efficient operations	3	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of Binary Search Tree operations. Implementing search and deletion operations and analyzing the time complexity.	3	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	3	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 1 Students come to know about AVL Tree and its operations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	3	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where continuous patient data insertion makes the system slow, by explaining the need for balancing	3	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of AVL Tree and need for balancing. Implementation of AVL insertion. Analyzing height and balance factor.	3	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	3	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 1 Students come to know about AVL Tree and its operations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	3	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, how to maintain balance after frequent updates, by explaining the concept of AVL Rotations	3	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of AVL rotations. Implementing insertion and deletion using rotations and comparing performance with Binary Search Tree.	3	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	3	Chalk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

SESSION NUMBER : 6

Session Outcome: 1 Students come to know about B Tree and its operations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	4	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where large patient data is stored in external memory, by explaining efficient indexing	4	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept and properties of B-Tree. Implementing search, insertion and deletion operations and analyzing height and efficiency	4	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

Session Outcome: 1 Students come to know about B Tree and its operations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	4	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, how to retrieve patient records within a range	4	Chalk	Quiz/Test Questions
30	Lecture: Discuss about concept of B+ Tree. Implementation of search and range queries. Analyzing advantages over B-Tree	4	Chalk	Quiz/Test Questions
5	Conclusion and Summary	4	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 1 Students come to know about Segment Tree and its operations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	4	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, how to find total number of patients in a given time interval, by explaining Segment Tree for range queries.	4	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of Segment Tree and its motivation. Implementing build operation, range query and	4	Chalk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
	point update and explaining lazy propagation technique. Discuss about concept of Fenwick Tree including Binary Indexed Tree indexing trick and comparing Segment Tree and Fenwick Tree.			---
5	Conclusion and Summary	4	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 1 Students come to know about Binary Indexed Tree and its operations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	4	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, how to efficiently update and query patient count dynamically, by explaining Binary Indexed Tree (Fenwick Tree).	4	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of Fenwick Tree. Implementing Binary Indexed Tree and solving inversion count problem and analyzing time complexity.	4	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 1 Students come to know about Graphs

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	3	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where hospitals are connected through referral systems and patient transfers between multiple locations need to be managed efficiently	3	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of graphs and terminology. Implementing graph representations using adjacency matrix and adjacency list and analyzing space and time complexity	3	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	3	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 1 Students come to know about Breadth First Search Algorithm

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	3	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where there is a need to find the nearest available hospital for emergency cases in minimum time	3	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of Breadth First Search. Implementing BFS using queue and analyzing time and space complexity and applications	3	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	3	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 1 Students come to know about Depth First Search Algorithm

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	3	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where all possible referral paths need to be explored to reach specialized treatment centers	3	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of Depth First Search. Implementing DFS using recursion and stack and analyzing traversal and connected components	3	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	3	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

Session Outcome: 1 Students come to know about Minimum cost Spanning Trees

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	3	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where hospitals need to be connected with minimum infrastructure cost	3	Chalk	Quiz/Test Questions
30	Lecture: Discuss about concept of Minimum Spanning Tree including definition, cut property and cycle property. Implementing Kruskal's Algorithm and Prim's Algorithm and analyzing their working. Comparison between Kruskal's (edge-centric, suitable for sparse graphs) and Prim's (vertex-centric, suitable for dense graphs).	3	Chalk	Quiz/Test Questions
5	Conclusion and Summary	3	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 14

Session Outcome: 1 Students come to know about Shortest Path Algorithms

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	4	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where shortest route is required for ambulance services to reach hospitals quickly.	4	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of shortest path problem. Implementing Dijkstra algorithm for single source shortest path(SSSP) with non-negative weights and analyzing time complexity	4	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

Session Outcome: 1 Students come to know about Priority Queue

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	4	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where emergency patients need to be handled based on severity and priority.	4	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of Greedy approach and its role in optimization problems. Implementing Priority Queue and applying it for shortest path computation including path reconstruction. Analyzing correctness proof of greedy strategy.	4	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

Session Outcome: 1 Students come to know about Shortest path algorithms

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	4	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where routes may have delays or varying conditions affecting travel time	4	Chalk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
30	Lecture: Discuss about concept of Bellman-Ford algorithm. Implementing shortest path computation including relaxation process for all edges and handling negative weights. Analyzing negative cycle detection and time complexity.	4	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	4	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 17

Session Outcome: 1 Students come to know about Shortest path algorithms

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	5	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where shortest paths between all hospitals are required for better coordination.	5	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of Floyd-Warshall algorithm. Implementing all-pairs shortest path using dynamic programming approach and updating distance matrix. Analyzing time complexity	5	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

Session Outcome: 1 Students come to know about Divide and Conquer

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	5	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where large patient data from multiple hospitals needs to be processed and sorted efficiently by dividing the data into smaller parts.	5	Chalk	Quiz/Test Questions
30	Lecture: Discuss about concept of Divide and Conquer strategy. Implementing Merge Sort and Quick Sort including partitioning and merging process. Analyzing best, average and worst case time complexity.	5	Chalk	Quiz/Test Questions
5	Conclusion and Summary	5	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 19

Session Outcome: 1 Students come to know about Sorting Techniques

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	5	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where hospitals or patients need to be ranked based on performance or priority for better decision making	5	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of advanced sorting techniques. Implementing Heap Sort and Quick Sort and analyzing performance for different input cases.	5	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

Session Outcome: 1 Students come to know about Sortings

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	5	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where large numerical data such as patient IDs need to be sorted faster for quick access	5	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of non-comparison sorting techniques. Implementing Counting Sort and Radix Sort and analyzing time complexity and limitations	5	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 21

Session Outcome: 1 Students come to know about Sortings

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	5	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where different methods are used to process patient data and there is a need to compare their performance	5	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of empirical benchmarking of algorithms. Implementing all six sorting algorithms and running them on different datasets including random, nearly sorted and reverse-sorted with input sizes 10K, 100K and 1M. Analyzing performance by filling benchmark table with algorithm, dataset, time and memory. Comparing results and justifying the best sorting technique for E-Commerce ranking system based on evidence	5	Chalk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Conclusion and Summary	5	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 22

Session Outcome: 1 Students come to know about Greedy Approach

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	5	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where limited resources like doctors or beds need to be used efficiently to treat maximum patients	5	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of Greedy approach and its properties. Implementing Activity Selection and Fractional Knapsack algorithms and analyzing correctness and efficiency of greedy strategy	5	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

Session Outcome: 1 Students come to know about Greedy Approach problems

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	5	Chalk	--- NOT APPLICABLE ---
10	Discuss about MediFlow, where medical records need to be stored in less space to manage large data efficiently.	5	Chalk	--- NOT APPLICABLE ---
30	Lecture: Discuss about concept of Huffman Coding and greedy strategy for compression. Implementing encoding and decoding process using Huffman Tree and analyzing compression efficiency.	5	Chalk	--- NOT APPLICABLE ---
5	Conclusion and Summary	5	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

Session Outcome: 1 Students come to know about Dynamic Programming

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Recap	5	Chalk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

10	Discuss about MediFlow, where repeated calculations in treatment planning need to be avoided to save time, Discuss about MediFlow, where best treatment plan needs to be selected under constraints like cost and resources.	5	Chalk	Quiz/Test Questions
30	Lecture: Discuss about concept of Dynamic Programming including overlapping subproblems and optimal substructure. Implementing memorization and tabulation techniques and analyzing efficiency, Discuss about concept of Dynamic Programming applications. Implementing 0/1 Knapsack, Longest Common Subsequence, Longest Increasing Subsequence and Matrix Chain Multiplication and analyzing time complexity and optimal solutions	5	Chalk	Quiz/Test Questions
5	Conclusion and Summary	5	Chalk	--- NOT APPLICABLE ---

Tutorial Course DELIVERY Plan:

NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Construct a Binary Search Tree (BST) by inserting a given set of elements and perform search operations for both existing and non-existing keys. Analyze the correctness of the tree structure	CO5
2	Implement an AVL Tree and demonstrate all types of rotations (LL, RR, LR, RL) during insertion. Verify the height balance property and compare its efficiency with a BST	CO5
3	Design and implement a B-Tree of order 5, insert a large set of keys, and verify that the height of the tree satisfies the expected constraints	CO5
4	Implement a B+ Tree with a range query operation and a Segment Tree with lazy propagation, and demonstrate their efficiency in handling range-based queries	CO5
5	Implement the Breadth-First Search (BFS) algorithm on a graph and display the traversal order along with the distance of each node from a given source vertex	CO5
6	Implement Depth-First Search (DFS) with cycle detection, and construct a Minimum Spanning Tree using Kruskal's algorithm with Union-Find, showing the selected edges and total cost	CO5
7	Implement Prim's algorithm for Minimum Spanning Tree and compare its total cost and performance with Kruskal's algorithm on graphs of varying sizes	CO5

Tutorial Session no	Topics	CO-Mapping
8	Implement Dijkstra's and Bellman-Ford algorithms and compare their results on weighted graphs, highlighting differences in handling edge weights	CO5
9	Implement the Floyd-Warshall algorithm to compute all-pairs shortest paths and perform topological sorting using Kahn's algorithm on a directed acyclic graph	CO5
10	Implement and analyze sorting algorithms such as Quick Sort, Merge Sort, and Heap Sort by evaluating their performance on different types of input datasets	CO5
11	Implement Counting Sort and Radix Sort algorithms, and demonstrate the step-by-step execution of Radix Sort on the dataset [170, 45, 75, 90, 802, 24, 2, 66] to illustrate digit-wise sorting	CO5
12	Solve optimization problems using Greedy and Dynamic Programming techniques, including Activity Selection, 0/1 Knapsack, Longest Common Subsequence (LCS), Matrix Chain Multiplication, and Edit Distance, and analyze their applicability	CO5

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Binary Search Tree (BST)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Construct a Binary Search Tree (BST) by inserting a given set of elements and perform search operations for both existing and non-existing keys	3	Chalk	--- NOT APPLICABLE ---
30	Analyze the correctness of the tree structure.	4	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 AVL Trees

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement an AVL Tree and demonstrate all types of rotations (LL, RR, LR, RL) during insertion.	3	Chalk	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
30	Verify the height balance property and compare its efficiency with a BST	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 1 B-TREE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Design and implement a B-Tree of order 5, insert a large set of keys	3	Chalk	--- NOT APPLICABLE ---
30	verify that the height of the tree satisfies the expected constraints.	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 1 B+ TREES

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement a B+ Tree with a range query operation and a Segment Tree with lazy propagation	3	Chalk	--- NOT APPLICABLE ---
30	Demonstrate their efficiency in handling range-based queries	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 1 Breadth-First Search (BFS) algorithm

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement the Breadth-First Search (BFS) algorithm on a graph	3	Chalk	--- NOT APPLICABLE ---
30	Display the traversal order along with the Distance of each node from a given source vertex	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 1 Depth-First Search (DFS)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement Depth-First Search (DFS) with cycle detection, and construct a Minimum Spanning Tree using Kruskal's algorithm	3	Chalk	--- NOT APPLICABLE ---
30	With Union-Find, showing the selected edges and total cost	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

Session Outcome: 1 Prim's algorithm

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement Prim's algorithm for Minimum Spanning Tree	3	Chalk	--- NOT APPLICABLE ---
30	Compare its total cost and performance with Kruskal's algorithm on graphs of varying sizes	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 1 Dijkstra's and Bellman-Ford algorithms

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement Dijkstra's and Bellman-Ford algorithms	3	Chalk	--- NOT APPLICABLE ---
30	compare their results on weighted graphs, highlighting differences in handling edge weights	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 1 Floyd-Warshall algorithm

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement the Floyd-Warshall algorithm to compute all-pairs shortest paths	3	Chalk	--- NOT APPLICABLE ---
30	perform topological sorting using Kahn's algorithm on a directed acyclic graph	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 1 Analyze sorting algorithms

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement and analyze sorting algorithms such as Quick Sort, Merge Sort, and Heap Sort	3	Chalk	--- NOT APPLICABLE ---
30	Evaluating their performance on different types of input datasets	4	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 1 Implement Counting Sort and Radix Sort algorithms

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement Counting Sort and Radix Sort algorithms, and demonstrate the step-by-step execution of Radix Sort on the dataset [170, 45, 75, 90, 802, 24, 2, 66]	3	Chalk	--- NOT APPLICABLE ---
30	Illustrate digit-wise sorting	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 1 Greedy and Dynamic Programming techniques

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Solve optimization problems using Greedy and Dynamic Programming techniques, including Activity Selection, 0/1 Knapsack, Longest Common Subsequence (LCS)	3	Chalk	--- NOT APPLICABLE ---
30	Matrix Chain Multiplication, and Edit Distance, and analyze their applicability	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

Skilling Course DELIVERY Plan:

Skilling session no	Topics/Experiments	CO-Mapping
1	Design and implement a data model in Java to represent the core entities of your selected project. Create appropriate classes and store records using a suitable structure	CO6
2	Implement Binary Search Tree (BST) in Java to manage project records efficiently. Perform insertion and traversal operations	CO6
3	Extend the BST implementation to support search and deletion operations, and analyze their performance in your project context	CO6
4	Implement a self-balancing AVL Tree in Java to handle frequent updates in your project and maintain optimal performance	CO6

Skilling session no	Topics/Experiments	CO-Mapping
5	Enhance the AVL Tree by implementing rotations and deletion, and compare its performance with BST for your project data	CO6
6	Implement a B-Tree (order-based) in Java to handle large-scale project data and demonstrate efficient indexing	CO6
7	Implement a B+ Tree with range query support and demonstrate how your project benefits from efficient range-based retrieval	CO6
8	XImplement Segment Tree (with lazy propagation) or Fenwick Tree (BIT) in Java to handle dynamic range queries in your project	CO6
9	Model your project as a graph and implement BFS and DFS in Java to explore connectivity and traversal	CO6
10	Implement Minimum Spanning Tree algorithms (Kruskal's and Prim's) in Java to optimize network/resource connections in your project	CO6
11	Implement Shortest Path algorithms (Dijkstra and Bellman-Ford) in Java to compute optimal paths in your project scenario	CO6
12	Implement sorting (Merge/Quick/Heap) and optimization techniques (Greedy & Dynamic Programming) in Java to solve real-world problems in your project	CO6

Skilling Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Create appropriate classes and store records using a suitable structure

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Design and implement a data model in Java to represent the core entities of your selected project	3	Chalk	--- NOT APPLICABLE ---
30	Create appropriate classes and store records using a suitable structure	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 1 Binary Search Tree (BST)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement Binary Search Tree (BST) in Java to manage project records efficiently	3	Chalk	--- NOT APPLICABLE ---
30	Perform insertion and traversal operations	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 1 BST implementation

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Extend the BST implementation to support search and deletion operations,	3	Chalk	--- NOT APPLICABLE ---
30	Analyze their performance in your project context	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 1 AVL Tree

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement a self-balancing AVL Tree in Java to handle frequent updates	3	Chalk	--- NOT APPLICABLE ---
30	In your project and maintain optimal performance	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 1 Enhance the AVL Tree by implementing rotations and deletion

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Enhance the AVL Tree by implementing rotations and deletion	3	Chalk	--- NOT APPLICABLE ---
30	Compare its performance with BST for your project data	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 1 B-Tree (order-based)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement a B-Tree (order-based) in Java to handle large-scale project data	3	Chalk	--- NOT APPLICABLE ---
30	Demonstrate efficient indexing.	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 7

Session Outcome: 1 B+ Tree

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement a B+ Tree with range query support	3	Chalk	--- NOT APPLICABLE ---
30	demonstrate how your project benefits from efficient range-based retrieval	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 1 Implement Segment Tree (with lazy propagation) or Fenwick Tree (BIT)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement Segment Tree (with lazy propagation) or Fenwick Tree (BIT) in Java to handle dynamic range queries in your project	3	Chalk	--- NOT APPLICABLE ---
30	Dynamic range queries in your project.	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 1 implement BFS and DFS

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Model your project as a graph and implement BFS and DFS in Java	3	Chalk	--- NOT APPLICABLE ---
30	Explore connectivity and traversal	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 1 Minimum Spanning Tree algorithms (Kruskal's and Prim's)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement Minimum Spanning Tree algorithms (Kruskal's and Prim's) in Java	3	Chalk	--- NOT APPLICABLE ---
30	optimize network/resource connections in your project	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

SESSION NUMBER : 11

Session Outcome: 1 Shortest Path algorithms (Dijkstra and Bellman-Ford)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement Shortest Path algorithms (Dijkstra and Bellman-Ford) in Java	3	Chalk	--- NOT APPLICABLE ---
30	compute optimal paths in your project scenario	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 1 sorting (Merge/Quick/Heap)

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	ATTENDANCE	1	Talk	--- NOT APPLICABLE ---
60	Implement sorting (Merge/Quick/Heap) and optimization techniques (Greedy & Dynamic Programming) in Java	3	Chalk	--- NOT APPLICABLE ---
30	solve real-world problems in your project	3	Chalk	--- NOT APPLICABLE ---
5	VIVA	2	Talk	--- NOT APPLICABLE ---

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
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COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									

	Hour	1	2	3	4	5	6	7	8	9
Mon	Theory	--	--	---	---	--	--	--	--	--
	Tutorial	--	--	---	---	--	--	--	--	--
	Lab	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	---	---	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9,V-S10,V-S10	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9,V-S10,V-S10	--	--	--
	Skilling	--	--	---	---	--	--	--	--	--
Tue	Theory	--	--	---	---	--	--	--	--	--
	Tutorial	--	--	---	---	--	--	--	--	--
	Lab	--	--	---	---	--	--	--	--	--
	Skilling	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	---	---	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9,V-S10,V-S10	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9,V-S10,V-S10	--	--	--
Wed	Theory	---	---	--	--	---	---	--	--	--
	Tutorial	---	---	--	--	---	---	--	--	--
	Lab	---	---	--	--	---	---	--	--	--
	Skilling	---	---	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9,V-S10,V-S10	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9,V-S10,V-S10	---	---	--	--	--
Thu	Theory	---	---	--	--	---	---	--	--	--
	Tutorial	---	---	--	--	---	---	--	--	--
	Lab	---	---	--	--	---	---	--	--	--
	Skilling	---	---	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	---	---	--	--	--
Fri	Theory	--	--	---	---	--	--	--	--	--
	Tutorial	--	--	---	---	--	--	--	--	--
	Lab	--	--	---	---	--	--	--	--	--
	Skilling	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9,V-S10,V-S10	V-S6,V-S6,V-S7,V-S7,V-S8,V-S8,V-S9,V-S9,V-S10,V-S10	---	---	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	V-S1,V-S1,V-S2,V-S2,V-S3,V-S3,V-S4,V-S4,V-S5,V-S5	--	--	--

	Hour	1	2	3	4	5	6	7	8	9
Sat	Theory	V-S6,V-S7,V-S8,V-S9,V-S10	V-S6,V-S7,V-S8,V-S9,V-S10	---	---	V-S1,V-S2,V-S3,V-S4,V-S5	V-S1,V-S2,V-S3,V-S4,V-S5	--	--	--
	Tutorial	--	--	---	---	--	--	--	--	--
	Lab	--	--	---	---	--	--	--	--	--
	Skilling	--	--	---	---	--	--	--	--	--
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplement course handout, which may perhaps include special lectures and discussions that would be planned, and schedule notified according

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4	CO5	CO6
End Semester Summative Evaluation Total= 40 %	Hackathon-Final Review	Weightage	20		180	3	3	3	3	4	4
		Max Marks	120			20	20	20	20	20	20
	End Semester Exam	Weightage	20		180	5	5	5	5		
		Max Marks	100			25	25	25	25		
In Semester Formative Evaluation Total= 35 %	Global Challenges	Weightage	10		90	2.5	2.5	2.5	2.5		
		Max Marks	120			30	30	30	30		
	Continuous Evaluation - Project	Weightage	10		90						10
		Max Marks	120								120
	Continuous Evaluation - Lab Exercise	Weightage	5		90					5	
		Max Marks	120							120	
	ALM	Weightage	10		90	2.5	2.5	2.5	2.5		

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4	CO5	CO6
		Max Marks	120			30	30	30	30		
In Semester Summative Evaluation Total= 25 %	Semester in Exam-II	Weightage	12.5		90			6.25	6.25		
		Max Marks	50					25	25		
	Semester in Exam-I	Weightage	12.5		90	6.25	6.25				
		Max Marks	50			25	25				

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course. In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments.

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Supplement course handout, which may perhaps include special lectures and discussions

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Supplement course handout, which may perhaps include special lectures and discussions

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Shaik Razia	L	10-MA	-	-	-	-
Shaik Razia	P	10-B,10-A	-	-	-	-
Shaik Razia	S	10-B,10-A	-	-	-	-
Zeelan CMAK	L	1-MA	-	-	-	-
Zeelan CMAK	P	1-B,1-A	-	-	-	-
Zeelan CMAK	S	1-B,1-A	-	-	-	-
Sandeepkumar Sornapudi	L	2-MA	-	-	-	-
Sandeepkumar Sornapudi	P	2-A,2-B	-	-	-	-
Sandeepkumar Sornapudi	S	2-B,2-A	-	-	-	-
KRISHNA CHAITANYA GOGINENI	L	9-MA	-	-	-	-
KRISHNA CHAITANYA GOGINENI	P	9-B,9-A	-	-	-	-
KRISHNA CHAITANYA GOGINENI	S	9-B,9-A	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Jogendra munagala	L	5-MA	-	-	-	-
Jogendra munagala	P	5-A,5-B	-	-	-	-
Jogendra munagala	S	5-B,5-A	-	-	-	-
Pavan Ande	L	3-MA	-	-	-	-
Pavan Ande	P	3-A,3-B	-	-	-	-
Pavan Ande	S	3-A,3-B	-	-	-	-
Burada Sreedhar	L	4-MA	-	-	-	-
Burada Sreedhar	P	4-A,4-B	-	-	-	-
Burada Sreedhar	S	4-B,4-A	-	-	-	-
Anjana Akurathi	L	6-MA	-	-	-	-
Anjana Akurathi	P	6-A,6-B	-	-	-	-
Anjana Akurathi	S	6-A,6-B	-	-	-	-
Venkata Suresh Bhavanam	L	7-MA	-	-	-	-
Venkata Suresh Bhavanam	P	7-A,7-B	-	-	-	-
Venkata Suresh Bhavanam	S	7-A,7-B	-	-	-	-
surya chebrolu	L	8-MA	-	-	-	-
surya chebrolu	P	8-B,8-A	-	-	-	-
surya chebrolu	S	8-A,8-B	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes and carry the text book(s) or material(s) as prescribed by the Course Faculty to the class.

NOTICES

Most of the notices are available on the LMS platform.

All notices will be communicated through the institution email.

All notices concerning the course will be displayed on the respective Notice Boards.

Signature of COURSE COORDINATOR

(Zeelan Basha CMAK)(3846)(DBES-1)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of DBES-1

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS
(Sign with Office Seal)
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